Spectral Index Discovery and Novelty Assessment: An Exhaustive Evaluation of Algorithmic Band Combinations

1. Epistemological Framework and Analytical Scope

The systematic classification of land use and land cover (LULC) relies heavily on the mathematical transformation of multispectral satellite data into indices that isolate specific biophysical phenomena. For decades, the remote sensing community has depended on a foundational taxonomy of normalized bounded formulas, most notably the Normalized Difference Vegetation Index (NDVI), the Modified Normalized Difference Water Index (MNDWI), and the Soil Adjusted Vegetation Index (SAVI).[1] These indices were primarily developed through human intuition, laboratory spectrometry, and an understanding of the physical absorption and reflectance properties of terrestrial materials.[4] Recently, however, algorithmic heuristic searches over massive combinations of spectral bands have been deployed to identify new mathematical relationships that maximize class separability specifically for machine learning classifiers.[1]

The subject document under evaluation presents the empirical results of an algorithmic search evaluating 836 unique band combinations applied to a complex LULC dataset situated in Arunachal Pradesh, India.[1] The computational outputs yield three primary claims of distinct scientific novelty that challenge conventional geospatial practices:

1. The algorithmic discovery of "Product-Over-Sum" (POS) formulations utilizing exclusively visible spectrum bands, which achieved Analysis of Variance (ANOVA) F-statistics exceeding 12,000.[1]
2. The implementation of un-harmonized, cross-sensor band subtractions—specifically fusing Landsat-8 Operational Land Imager (OLI) and Sentinel-2 Multispectral Instrument (MSI) bands—to achieve the highest overall class separability in the dataset, surpassing an F-statistic of 13,000.[1]
3. The empirical observation that raw signal differences (un-normalized subtractions) are vastly superior to traditional normalized bounded indices when ingested by tree-based machine learning algorithms, thus rendering traditional normalization mathematically obsolete for such classifiers.[1]

This research report provides an exhaustive, peer-level deconstruction of these three core claims. By cross-referencing these algorithmic outputs against established physical principles of optical remote sensing, statistical mechanics, radiometric calibration protocols, and machine learning theory, this assessment meticulously determines the genuine novelty, mathematical validity, and practical, scalable utility of the discovered formulations. The analysis synthesizes current literature regarding multi-sensor harmonization, the mathematical mechanics of

recursive partitioning algorithms, and the profound topographical influence of the Himalayan terrain on spectral reflectance.

2. Evaluation of Product-Over-Sum (POS) Formulations

The algorithmic search identified three strictly single-sensor (Landsat-8) indices that discard shortwave infrared (SWIR) and near-infrared (NIR) dependence in favor of visible spectrum interactions, specifically utilizing a Product-Over-Sum structure.[1] The highest performing of these formulations, utilizing the blue and green bands, achieved an ANOVA F-statistic of 12,821, ranking third overall in the analyzed dataset.[1] The subsequent combinations, utilizing blue and red, and blue and NIR, also achieved extreme statistical significance.

2.1 Mathematical Origins Versus Remote Sensing Applications

To accurately ascertain the novelty of the POS formulation, one must separate its underlying mathematical origin from its domain-specific application in Earth Observation. Mathematically, the structural formula $\frac{A \times B}{A + B}$ is unequivocally established in the physical and computational sciences. It is universally recognized as the fundamental equation for calculating the equivalent resistance of two parallel resistors in electrical circuit theory.[8] Statistically, it represents exactly one-half of the harmonic mean of two variables. Furthermore, the structural property of the product divided by the sum appears extensively in discrete element analysis for calculating inter-species density [11] and in convex optimization functions.[12]

However, in the specific context of optical remote sensing and spectral index design, the application of the harmonic mean or the POS structure as a primary spatial feature is verifiably absent from standard algorithmic and programmatic index catalogues. Exhaustive reviews of established spectral databases, such as the IndexDataBase and VegSpecIndex, reveal a heavy historical reliance on normalized differences $\frac{A-B}{A+B}$, simple ratios $\frac{A}{B}$, un-normalized differences $A-B$, or linear combinations with atmospheric dampening constants (such as the Enhanced Vegetation Index or the Atmospherically Resistant Vegetation Index).[2]

While recent empirical research occasionally explores non-linear index combinations or complex multi-band derivations—such as the Blue colour-coated steel roof index (BCCSI) or the Normalized Difference Colourimetric High Resolution Water Index (NDCHRWI)—these modern indices still universally rely on traditional normalized difference ratios or additive linear combinations.[18] For instance, attempts to utilize the blue band for specific target detection typically involve variations of $(Blue - Green)/(Blue + Green)$ or complex band weighting, not multiplicative relationships.[21] Therefore, the explicit deployment of the POS formula ($\frac{Band1 \times Band2}{Band1 + Band2}$) as an independent feature engineered specifically to maximize class variance represents a genuine and significant methodological novelty in optical remote sensing feature engineering.

2.2 The Anchor Mechanism of the Visible Blue Band

The algorithmic search explicitly highlights the Landsat-8 Blue band (0.45–0.51 µm) as the "anchor band" for these novel POS formulations.[1] The claim that a POS index relying solely on the blue and green bands can achieve extreme separability (F-statistic 12,821) without utilizing the SWIR bands represents a substantial deviation from orthodox water and moisture extraction methodologies.[1] In traditional analyses, the SWIR bands (e.g., SWIR1 at 1.57-1.65 µm and SWIR2 at 2.08-2.35 µm) are considered mathematically indispensable for aquatic discrimination due to water's near-total absorption of electromagnetic radiation beyond the 1.2 µm threshold.[23]

The success of the visible-band POS formulation lies deeply embedded in its non-linear scaling properties and the unique interaction of visible light with distinct Earth surface materials. The formula organically functions as a continuous "logical AND" gate for high reflectance. To understand this mathematically, consider the behavior of the equation assuming top-of-atmosphere (TOA) or surface reflectance (SR) values bounded between 0 and 1:

- **High Dual Reflectance:** If both band A and band B are highly reflective (e.g., 0.8), the POS output maintains a relatively high magnitude ($\frac{0.64}{1.6} = 0.40$).
- **Low Dual Reflectance:** If both bands are deeply absorptive (e.g., 0.1), the POS output is exponentially suppressed ($\frac{0.01}{0.2} = 0.05$).
- **Asymmetrical Reflectance:** Crucially, if one band is highly reflective while the other is highly absorptive (e.g., $A = 0.8$, $B = 0.1$), the output is severely dampened toward the lower value ($\frac{0.08}{0.9} = 0.088$).

2.3 Biophysical Interpretation of the Scaling Properties

This non-linear scaling mechanism is uniquely tailored to the biophysical realities of the visible spectrum. In natural, un-built landscapes, healthy photosynthetic vegetation vigorously absorbs both blue and red electromagnetic radiation to fuel chlorophyll-a and chlorophyll-b interactions, while moderately reflecting green light.[4] Because the blue reflectance is inherently low, the asymmetrical reflectance rule applies, and the POS value is mathematically suppressed. Similarly, bare soils exhibit a distinct spectral signature characterized by a gradual linear increase in reflectance from the blue band through the red band, keeping the initial blue reflectance comparatively low and thus suppressing the overall POS output.

Conversely, specific targets exhibit simultaneous high reflectance in the shorter wavelengths. Shallow water, turbid aquatic environments carrying high suspended sediment loads, and highly specific built-up materials (such as blue colour-coated industrial roofing, high-albedo urban concrete, and certain artificial surfaces) reflect strongly and simultaneously in both the blue and green visible bands.[19] Furthermore, the blue band is highly sensitive to atmospheric scattering (Rayleigh scattering) and shallow water penetration, providing an excellent medium for detecting water bodies where deep SWIR absorption might otherwise confuse dark shadows with actual water.[23]

By utilizing the POS structure, the heuristic algorithm has organically discovered a mathematical filter that aggressively squashes the statistical variance of terrestrial, photosynthetic, and geologic classes into a dense, low-value cluster, while simultaneously stretching the variance of aquatic and high-albedo artificial targets across a much wider

numerical continuum. Therefore, the assertion that the POS structure is a novel and highly effective discovery for spatial classification is entirely validated by its underlying mathematical and biophysical mechanics.

Index Formulation	Wavelength Range (Landsat-8)	Primary Biophysical Interaction	Algorithmic Mechanism
PR(Blue, Green)	0.45µm - 0.59µm	Turbid water, shallow bathymetry, artificial surfaces	Amplifies concurrent short-wavelength reflectance
PR(Blue, Red)	0.45µm - 0.67µm	High-albedo barren land, severe sediment loads	Suppresses healthy vegetation (chlorophyll absorption)
PR(Blue, NIR)	0.45µm - 0.90µm	Vegetation structure vs. water absorption	Maximizes contrast between cellular structure and moisture

Table 1: Analysis of the discovered Product-Over-Sum (POS) formulations, detailing their respective wavelength ranges and the biophysical interactions driving the algorithmic separability.

3. The Un-Normalized Paradigm in Tree-Based Machine Learning

The document asserts a major paradigm shift for applied geospatial machine learning: that traditional bounding normalization (e.g., $\frac{A-B}{A+B}$) is entirely unnecessary for tree-based algorithms such as XGBoost, and that raw signal differences (e.g., $A - B$) yield mathematically stricter class separability.[1]

3.1 Bounded Ratios vs. Unbounded Variance in Recursive Partitioning

This claim represents a critical intersection of historical remote sensing tradition and modern computer science architecture. Within the domain of deep learning, artificial neural networks, and distance-based classifiers (such as K-Nearest Neighbors or Support Vector Machines), feature normalization or standardization (e.g., Min-Max scaling, Z-score normalization) is strictly required. This pre-processing step ensures uniform weight distribution, accelerates gradient descent convergence, prevents gradient explosion, and ensures that features with larger numerical ranges do not disproportionately dominate the loss function.[26]
However, the architecture of ensemble decision tree algorithms—including Classification and Regression Trees (CART), Random Forest, LightGBM, and eXtreme Gradient Boosting (XGBoost)—operates on entirely different mathematical principles. These recursive partitioning algorithms evaluate potential node splits by searching across the numerical distribution of a single feature to maximize Information Gain or minimize Gini Impurity.[29] Because they operate via orthogonal hyperplanes (axis-aligned splits), tree-based models are completely invariant to

monotonic transformations. The absolute scale of the data is irrelevant; the algorithm merely requires a sorted order of values to determine the optimal split threshold.[30]
Crucially, the transformation from a raw difference ($A - B$) to a normalized difference ($\frac{A-B}{A+B}$) is *not* a monotonic transformation. The denominator $(A + B)$ acts as a dynamic brightness normalizer, which fundamentally alters the rank order of the pixels within the dataset.

3.2 The Destruction of Albedo Magnitude Through Normalization

The remote sensing community has historically favored normalized bounded formulas specifically to suppress magnitude. Consider two hypothetical aquatic pixels observed by a multispectral sensor:

- **Pixel 1 (Deep, clear oceanic water):** Green Reflectance = 0.08, SWIR Reflectance = 0.02.
    - Raw Difference = $0.08 - 0.02 = 0.06$.
    - Normalized Difference = $\frac{0.06}{0.10} = 0.60$.
- **Pixel 2 (Shallow, highly reflective turbid river water):** Green Reflectance = 0.40, SWIR Reflectance = 0.10.
    - Raw Difference = $0.40 - 0.10 = 0.30$.
    - Normalized Difference = $\frac{0.30}{0.50} = 0.60$.

In the traditional normalized paradigm, both pixels evaluate to exactly 0.60. The normalization successfully suppresses the overall magnitude of the reflectance (the albedo), presenting both a dark ocean pixel and a bright river pixel as identical "water" signatures to a human visual interpreter or a linear thresholding model.
However, in the raw difference paradigm, the calculated values remain 0.06 and 0.30, respectively. The mathematical transformation preserves the underlying magnitude scalar. By discarding normalization, the un-normalized index $(A-B)$ provides the XGBoost algorithm with a multi-dimensional feature space that retains both the spectral contrast (the slope of the spectral signature) and the absolute intensity scalar (the brightness). The decision tree algorithm can therefore establish multiple conditional splits to separate classes that share identical spectral shapes but differ in absolute brightness.
The algorithmic search evaluated in the subject document has empirically proven that abandoning the human-centric requirement for bounded indices (scaled arbitrarily between -1 and +1) unlocks hidden variance for machine learning classifiers. While the concept that tree-based models do not require normalization is a known tenet in core computer science [27], the empirical proof that raw spectral subtractions outperform normalized indices in geospatial feature engineering serves as a profound and highly actionable applied insight for the remote sensing industry.

4. Topographical Confounding: The Arunachal Pradesh Context

The geographical and ecological context of the dataset—specifically the mountainous terrain

of Arunachal Pradesh, India—adds a critical, indispensable layer to understanding why un-normalized differences achieved such massive statistical scores. The document indicates that raw subtractions (such as $L8_{red} - L8_{swir1}$) act as structural equivalents to known indices but vastly outperform them in F-statistics.[1]

4.1 Rugged Terrain and the Bidirectional Reflectance Distribution Function (BRDF)

Arunachal Pradesh is characterized by extreme Himalayan topography, featuring severe slopes, deep river valleys, and highly rugged terrain.[33] This topographic complexity creates severe illumination differences across the landscape due to varying solar zenith angles and terrain aspects. A sun-facing slope will reflect vastly more solar energy to the satellite sensor than an identical land cover situated on a shadowed, north-facing slope.[34]

Historically, normalized indices were designed and adopted by the remote sensing community precisely to cancel out these illumination variations.[14] By dividing the difference of two bands by their sum, the normalized ratio largely mitigates the scalar effect of the topographic illumination, producing a relatively uniform index value for a continuous forest canopy regardless of whether that canopy is fully illuminated or in deep shadow.

4.2 Raw Subtraction as a Digital Elevation Model Proxy

By relying on a raw mathematical subtraction ($A-B$), the discovered index fundamentally fails to cancel out the illumination scalar. Instead, it encodes the topographic shadow map directly into the feature variable. In highly rugged terrain, specific land cover classes are intrinsically tied to specific topographies. For example, in Arunachal Pradesh, dense primary forests occupy the steep mountain slopes, while built-up settlements, agricultural lands, and river networks are restricted to the flatter, lower-elevation valleys.[34]

The massive F-statistics generated by the raw subtraction indices strongly suggest that the machine learning algorithm is effectively weaponizing the un-normalized topographic illumination variance to perfectly separate valley-bottom classes from mountain-slope classes. Because the raw subtraction preserves brightness, and brightness is dictated by topography, the raw subtraction serves as an implicit, high-resolution Digital Elevation Model (DEM) and slope proxy.

Therefore, while the algorithmic discovery is highly optimized for the Arunachal Pradesh dataset, its extreme statistical performance is partially an artifact of topographical confounding. If these raw subtraction formulas were applied to a perfectly flat, homogeneous landscape where illumination is uniform, the separability gap between the raw subtraction and the normalized index would likely narrow significantly.

5. The Mechanics and Validity of Cross-Sensor Band Subtraction

The most provocative and statistically extreme claim in the document concerns cross-sensor band fusion. The heuristic algorithm generated composite indices by subtracting raw

Sentinel-2 Shortwave Infrared bands directly from Landsat-8 visible bands, yielding the highest F-statistics in the entire dataset. The premier index, $L8_{blue} - S2_{SWIR1}$, achieved an F-statistic of 13,473, theoretically outperforming the benchmark MNDWI by a staggering factor of 3.17.[1]

5.1 The Established Paradigm of Multi-Sensor Harmonization

Multi-sensor data fusion is currently a massive focal point of advanced remote sensing research, predominantly driven by the desire to increase temporal resolution by combining the 16-day revisit cycle of the Landsat-8 satellite with the 5-day revisit cycle of the Sentinel-2 constellation.[39] However, the established scientific protocols for this fusion require intense radiometric calibration. Prominent initiatives, such as the Harmonized Landsat and Sentinel-2 (HLS) project, utilize complex, multi-stage pipelines to force the spectral response functions of the two distinct satellites into a comparable baseline.[41]

These harmonization efforts involve:

- **Atmospheric Correction:** Applying rigorous physical models, such as the Second Simulation of the Satellite Signal in the Solar Spectrum (6S), Fast Line-of-sight Atmospheric Analysis of Hypercubes (FLAASH), or Dark Object Subtraction (DOS), to convert Top-Of-Atmosphere (TOA) radiance to Surface Reflectance (SR).[41]
- **BRDF Normalization:** Adjusting for the differing view angles and solar illumination geometries between the two satellite passes.[46]
- **Spatial Resampling:** Reconciling the 30-meter resolution of Landsat-8 bands with the 10-meter (visible) and 20-meter (SWIR) resolutions of Sentinel-2.[47]
- **Spectral Bandpass Adjustment:** Utilizing Pseudo-Invariant Features (PIFS) to statistically correct for the slight differences in the wavelengths captured by the respective sensors.[44]

5.2 The Algorithmic Artifact of Un-Harmonized Fusion

The direct mathematical subtraction of an un-harmonized Sentinel-2 band from a Landsat-8 band, as presented in the document, is entirely unorthodox and mathematically perilous. Because the two satellites possess different spatial resolutions, different spectral response widths, and—most critically—rarely capture identical scenes simultaneously, a direct subtraction incorporates massive radiometric and temporal noise.[47]

While the document classifies this cross-sensor formula as a "novel discovery" equivalent to the MNDWI structure [1], the extreme statistical separability (F-statistic > 13,000) strongly implies the presence of a severe algorithmic artifact—specifically, temporal data leakage.

If the Landsat-8 and Sentinel-2 images used in the Arunachal Pradesh dataset were acquired on different dates (which is highly probable given orbit schedules), the algorithmic subtraction $L8_{blue} - S2_{SWIR1}$ is not merely calculating the spectral difference between blue and SWIR light. It is fundamentally calculating:

$$Index = Reflectance(Blue)_{T1} - Reflectance(SWIR)_{T2} \pm \Delta Atmosphere \pm \Delta Sensor Bias \pm \Delta Phenology$$

In the dynamic environment of Arunachal Pradesh, a temporal gap of even a few days between the Landsat and Sentinel acquisitions would capture changes in soil moisture, agricultural phenology, cloud shadow displacement, and river hydrology.[33] The machine learning classifier is not evaluating a pure spectral index; it is exploiting this multi-temporal Δ (delta) as a highly discriminative feature to separate classes. It functions similarly to an anomalous change detection algorithm, rather than a universal baseline spectral index.[50]

Therefore, while the cross-sensor formula is indisputably novel in its raw, unfiltered formulation, it cannot be classified as a universally applicable spectral index. It is an idiosyncratic, highly localized feature engineered by the algorithm that maximizes variance by exploiting the temporal and radiometric incongruities between two unsynchronized satellite passes. Practitioners attempting to utilize this exact cross-sensor subtraction formula on a perfectly coincident dataset, or a rigorously harmonized dataset, would likely observe a catastrophic collapse in the reported F-statistic, as the temporal data leakage would be eliminated.

Harmonization Step	**Requirement for Cross-Sensor Fusion**	**Impact of Omission in Raw Subtraction**
Spatial Resampling	Align 30m Landsat with 20m Sentinel SWIR	Introduces pixel-boundary artifacts and spatial aliasing
Atmospheric Alignment	Standardize TOA to Surface Reflectance	Embeds differing aerosol/haze loads into the index value
BRDF Correction	Normalize view/illumination angles	Encodes differing shadow lengths into the spectral difference
Temporal Synchronization	Ensure simultaneous ground observation	**Severe Data Leakage:** Creates an index based on temporal change

Table 2: Analysis of necessary harmonization steps for multi-sensor remote sensing, highlighting the structural flaws and algorithmic artifacts introduced by the direct cross-sensor subtraction methodology.

6. Statistical Rigor: Decoding the F-Statistic Extremes

To fully contextualize the novelty and validity of these findings, one must rigorously analyze the reported ANOVA F-statistics. The F-statistic is mathematically defined as the ratio of the between-class variance (the dispersion of the class means relative to the overall mean) to the within-class variance (the dispersion of the individual observations relative to their specific class means).[51]

$$F = \frac{\text{Mean Square Between (MSB)}}{\text{Mean Square Within (MSW)}}$$

In standard geospatial literature and environmental statistics, an F-statistic in the high hundreds or low thousands is generally considered indicative of excellent class separability, representing distinct populations with minimal overlap.[54] The baseline score of 4,245 reported

for the widely respected MNDWI confirms that the underlying dataset contains well-defined classes.[1]

However, reporting F-statistics exceeding 13,000 indicates a nearly absolute, mathematically orthogonal separation of hyperplanes within the multi-dimensional feature space. This represents an astronomical degree of variance capture.

6.1 The Dominance of Aquatic Discrimination

This 3x improvement over established baselines highlights the absolute dominance of water and moisture discrimination in driving algorithmic variance.[1] All of the top-performing structural equivalents identified by the search—ranging from the Moisture Stress Index (MSI) to the New Water Index (NWI) and the Automated Water Extraction Index (AWEI)—are fundamentally moisture or water-sensitive mechanisms.[1]

Water exhibits the most extreme spectral divergence of any natural land cover; its near-total absorption of SWIR energy, juxtaposed with moderate visible reflectance, provides a mathematically perfect anchor point for maximizing variance across a dataset.[24] The algorithmic search relentlessly prioritized formulas that pushed the aquatic cluster as far away from the terrestrial and built-up clusters as mathematically possible.

The extreme F-statistics of 13,473 for the un-harmonized cross-sensor fusion, and 12,821 for the POS visible-band formulation, prove that the traditional practice of bounding indices compresses variance unnecessarily. By allowing values to expand uninhibited along the number line, the heuristic search maximized the numerator (MSB) of the F-statistic equation. The algorithm is unconcerned with the human interpretability of the index value (e.g., establishing a 0.0 threshold for water); it is exclusively optimized to maximize the distance between the statistical centroids of the targeted classes.

7. Strategic Implications for Earth Observation and Feature Engineering

The empirical results documented in the Arunachal Pradesh algorithmic study signify a critical inflection point in spatial data science and remote sensing methodology. Historically, spectral indices were painstakingly developed through physical intuition, laboratory spectrometry, and extensive field validation (e.g., identifying the exact absorption trough of chlorophyll-a or the precise reflectance peak of bare soil).[4] These biophysically sound indices were subsequently hard-coded into classification workflows and expected to perform universally across diverse geographies.

The algorithmic search deployed in the subject document represents the precise inverse of this historical approach: it utilizes a brute-force, empirical optimization architecture that treats spectral bands not as rigid physical phenomena, but merely as abstract numerical vectors in a high-dimensional feature space. The machine learning model does not inherently "know" what a blue band or a shortwave infrared band represents in the real world; it only knows which mathematical permutations create the widest standard deviations between labeled classes during the training phase.

This purely empirical approach forces practitioners to critically re-evaluate the utility of legacy indices. If a non-linear tree-based algorithm can extract triple the statistical variance from a raw, un-normalized subtraction than from a meticulously bounded normalized formula, then the standard practice of feeding pre-calculated NDVI or MNDWI features into an XGBoost model is computationally sub-optimal. The algorithm is inherently more capable of defining its own non-linear interaction terms and establishing its own variance-maximizing hyperplanes during the training phase.

The discovery of the Product-Over-Sum mechanism underscores this reality; an algorithm unconstrained by human logic regarding "normalized differences" will inevitably identify alternative mathematical pathways—such as the harmonic mean—to isolate target classes with superior efficacy.

8. Synthesized Conclusions on Index Novelty

Based on a comprehensive integration of physical remote sensing principles, complex statistical theory, the mathematics of recursive partitioning, and the specific topographical realities of the dataset, the novelty and validity of the claims presented within the subject document are evaluated as follows:

1. **Product-Over-Sum (POS) Formulas**: **Genuinely Novel in Domain Application.** While mathematically identical to parallel resistance physics and discrete element analysis formulas, the explicit deployment of $\frac{Blue \times Green}{Blue + Green}$ as a primary spectral index for LULC classification is a novel and highly effective discovery. It functions as a non-linear thresholding gate that maximizes the variance of high-albedo and aquatic targets while aggressively suppressing the variance of terrestrial vegetation and soil, rooted securely in the biophysical properties of visible light.
2. **Cross-Sensor Un-Normalized Subtraction**: **Novel in Formulation, but Methodologically Flawed.** Fusing un-harmonized Landsat-8 and Sentinel-2 bands via direct mathematical subtraction is unprecedented in standard literature, but it relies heavily on an algorithmic artifact. The extreme F-statistic (13,473) is virtually guaranteed to be a product of temporal data leakage, wherein the algorithm is exploiting the temporal, atmospheric, and radiometric deltas between two asynchronous satellite passes rather than measuring a stable, genuine surface property. It is an index of change, not a universal spectral baseline.
3. **Un-Normalized Subtractions for Machine Learning**: **An Applied Paradigm Shift.** The assertion that normalization is unnecessary is a standard tenet of fundamental data science and decision tree architecture. However, the empirical demonstration that raw spectral subtractions capture intensity magnitudes and implicit topographic proxies—thereby yielding triple the class separability (F-statistics) of standard bounded indices—is a highly valuable insight that challenges decades of remote sensing tradition regarding feature pre-processing.

Ultimately, the document successfully proves that algorithmic heuristic searches can vastly outperform human-engineered spectral indices in optimizing data for machine learning ingestion. It demonstrates unequivocally that the future of Earth Observation classification lies

not in generating new bounded, human-interpretable acronyms, but in supplying raw, un-normalized multidimensional data to algorithmic architectures capable of organically discovering the most variance-rich geometric projections.

Works cited

1. spectral_index_novelty_analysis.txt
2. IDB - Index: Normalized Difference NIR/Blue Blue-normalized difference vegetation index, accessed on March 27, 2026, https://www.indexdatabase.de/db/i-single.php?id=135
3. VARIABILITY OF REMOTE SENSING SPECTRAL INDICES IN BOREAL LAKE BASINS, accessed on March 27, 2026, https://isprs-archives.copernicus.org/articles/XLII-2/411/2018/
4. Blue NIR+Red Blue Red NIR NDVI= –, accessed on March 27, 2026, https://www.iitf.lbtu.lv/conference/proceedings2019/Papers/N465.pdf
5. Rapid Quantification of Microalgae Growth with Hyperspectral Camera and Vegetation Indices - JYX: JYU, accessed on March 27, 2026, https://jyx.jyu.fi/jyx/Record/jyx_123456789_74237
6. Sensors, Volume 26, Issue 3 (February-1 2026) – 328 articles - MDPI, accessed on March 27, 2026, https://www.mdpi.com/1424-8220/26/3
7. Attribute Filtering in Approximate Nearest Neighbor Search: An In-depth Experimental Study, accessed on March 27, 2026, https://arxiv.org/html/2508.16263v1
8. Electronic and Electrical Engineering - RPITST, accessed on March 27, 2026, https://rpitst.com/img/ebook/1711064928_467adda18af93a9d3941.pdf
9. Basics of Circuits - DOKUMEN.PUB, accessed on March 27, 2026, https://dokumen.pub/basics-of-circuits.html
10. ELECTRONIC PRINCIPLES, accessed on March 27, 2026, http://103.203.175.90:81/fdScript/RootOfEBooks/E%20Book%20collection%20-%202022/ECE/Malvino_Electronic-Principles%208th.pdf
11. Elasticity and Wave Propagation in Granular Materials, accessed on March 27, 2026, https://www2.msm.ctw.utwente.nl/sluding/THESIS/PhD_Taghizadeh.pdf
12. Convex Optimization - Stanford University, accessed on March 27, 2026, https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf
13. Awesome Spectral Indices for Google Earth Engine, accessed on March 27, 2026, https://awesome-ee-spectral-indices.readthedocs.io/_/downloads/en/latest/pdf/
14. VegSpecIndex: Vegetation and Spectral Indices for Environmental Assessment - CRAN, accessed on March 27, 2026, https://cran.rstudio.com/web/packages/VegSpecIndex/VegSpecIndex.pdf
15. Evaluation and Selection of Multi-Spectral Indices to Classify Vegetation Using Multivariate Functional Principal Component Analysis - iris univpm, accessed on March 27, 2026, https://iris.univpm.it/retrieve/fd4f539a-b30b-49c6-bab7-17d9ecd8871b/remotesensing-16-01224-v2.pdf
16. Supplementary material – table of concentration - BG, accessed on March 27,

2026, https://bg.copernicus.org/preprints/bg-2021-250/bg-2021-250-supplement.pdf
17. Package 'VegSpecIndex' reference manual - cran, accessed on March 27, 2026, https://cran.r-universe.dev/VegSpecIndex/doc/manual.html
18. Unmanned aerial vehicle for the assessment of woody and herbaceous phytomass in Sahelian savanna - Horizon IRD, accessed on March 27, 2026, https://horizon.documentation.ird.fr/exl-doc/pleins_textes/2022-02/010084079.pdf
19. Full article: A novel spectral index for mapping blue colour-coated steel roofs (BCCSRs) in urban areas using Sentinel-2 data - Taylor & Francis, accessed on March 27, 2026, https://www.tandfonline.com/doi/full/10.1080/17538947.2023.2241427
20. Holistic Reduction to Compare and Create New Indices for Global Inter-Seasonal Monitoring: Case Study for High Resolution Surface Water Mapping - MDPI, accessed on March 27, 2026, https://www.mdpi.com/2072-4292/15/8/2063
21. Potentiality of SDGSAT-1 glimmer imagery to investigate the spatial variability in nighttime lights, accessed on March 27, 2026, https://ira.lib.polyu.edu.hk/bitstream/10397/102179/1/1-s2.0-S1569843223001358-main.pdf
22. MapBiomas Trinational Pampa, accessed on March 27, 2026, https://pampa.mapbiomas.org/wp-content/uploads/sites/9/2023/11/ATBD-Collection-1_PampaTrinational.pdf
23. Spectral Signature Cheatsheet in Remote Sensing - GIS Geography, accessed on March 27, 2026, https://gisgeography.com/spectral-signature/
24. Automatic extraction of glacial lakes from Landsat imagery using deep learning across the Third Pole region, accessed on March 27, 2026, https://elib.dlr.de/206369/1/1-s2.0-S0034425724004395-main.pdf
25. A novel spectral index for mapping blue colour-coated steel roofs (BCCSRs) in urban areas using Sentinel-2 data - Taylor & Francis, accessed on March 27, 2026, https://www.tandfonline.com/doi/pdf/10.1080/17538947.2023.2241427
26. Identification of maize kernel varieties based on interpretable ensemble algorithms - PMC, accessed on March 27, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11927534/
27. A Comparative Analysis to Visualize the Behavior of Different Machine Learning Algorithms for Normalized and Un-Normalized Data in Predicting Alzheimer's Disease - ResearchGate, accessed on March 27, 2026, https://www.researchgate.net/publication/336738763_A_Comparative_Analysis_to_Visualize_the_Behavior_of_Different_Machine_Learning_Algorithms_for_Normalized_and_Un-Normalized_Data_in_Predicting_Alzheimer's_Disease
28. Identification of maize kernel varieties based on interpretable ensemble algorithms - Frontiers, accessed on March 27, 2026, https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2025.1511097/pdf
29. Optimized Deep Learning Model for Predicting Liver Metastasis in Colorectal Cancer Patients - MDPI, accessed on March 27, 2026,

https://www.mdpi.com/2073-8994/17/1/103
30. A new machine learning method for rainfall classification: temporal random tree - PMC, accessed on March 27, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12453857/
31. Enhancing Credit Risk Prediction in Retail Banking: Integrating Time Series and Classical ML Algorithms, accessed on March 27, 2026, https://essay.utwente.nl/fileshare/file/98283/Goldmann_MA_BMS1.pdf
32. Floodplain Lake Water Level Prediction with Strong River-Lake Interaction Using the Ensemble Learning LightGBM | Request PDF - ResearchGate, accessed on March 27, 2026, https://www.researchgate.net/publication/381409801_Floodplain_Lake_Water_Level_Prediction_with_Strong_River-Lake_Interaction_Using_the_Ensemble_Learning_LightGBM
33. arunachal pradesh - NATIONAL WETLAND ATLAS: - Space Applications Centre ISRO, accessed on March 27, 2026, https://vedas.sac.gov.in/content/vcms/downloads/atlas/Wetlands/NWIA_ArunachalPradesh_Atlas.pdf
34. Full article: Remote-sensing and GIS-based landslide-susceptibility zonation using the landslide index method in Igo River Basin, Eastern Himalaya, India - Taylor & Francis, accessed on March 27, 2026, https://www.tandfonline.com/doi/full/10.1080/01431161.2011.633121
35. the indian ecological society, accessed on March 27, 2026, https://indianecologicalsociety.com/wp-content/themes/ecology/fullpdfs/1625815547.pdf
36. ISSN (Online) - Journal of Wildlife Science (JWLS), accessed on March 27, 2026, https://jwls.in/wp-content/uploads/2024/12/Full-PDF_JWLS.SOBB1058.pdf
37. (PDF) Land Use and Land Cover Change Study Using RS-GIS and Its Impact on Biodiversity: Study from Arunachal Pradesh, India - ResearchGate, accessed on March 27, 2026, https://www.researchgate.net/publication/376780540_Land_Use_and_Land_Cover_Change_Study_Using_RS-GIS_and_Its_Impact_on_Biodiversity_Study_from_Arunachal_Pradesh_India
38. Full article: Spatial and temporal dynamic of land-cover/land-use and carbon stocks in Eastern Cameroon: a case study of the teaching and research forest of the University of Dschang - Taylor & Francis, accessed on March 27, 2026, https://www.tandfonline.com/doi/full/10.1080/21580103.2018.1520743
39. Landsat 8 and Sentinel-2 Fused Dataset for High Spatial-Temporal Resolution Monitoring of Farmland in China's Diverse Latitudes - MDPI, accessed on March 27, 2026, https://www.mdpi.com/2072-4292/15/11/2951
40. multi-sensor crop type classification using pixel- set encoders and temporal self-attention - Paris Lodron Universität Salzburg, accessed on March 27, 2026, https://eplus.uni-salzburg.at/obvusbhs/content/titleinfo/6915570/full.pdf
41. Sentinel-2 and Landsat-8 Observations for Harmful Algae Blooms in a Small Eutrophic Lake, accessed on March 27, 2026, https://www.mdpi.com/2072-4292/13/21/4479

42. An improved dark-object subtraction technique for atmospheric correction of Landsat 8 | Request PDF - ResearchGate, accessed on March 27, 2026, https://www.researchgate.net/publication/314596635_An_improved_dark-object_subtraction_technique_for_atmospheric_correction_of_Landsat_8
43. Effect of Atmospheric Corrections on NDVI: Intercomparability of Landsat 8, Sentinel-2, and UAV Sensors - ResearchGate, accessed on March 27, 2026, https://www.researchgate.net/publication/354414168_Effect_of_Atmospheric_Corrections_on_NDVI_Intercomparability_of_Landsat_8_Sentinel-2_and_UAV_Sensors
44. RADIOMETRIC CALIBRATION OF LANDSAT-8 OLI AND SENTINEL-2 MSI IMAGES FOR WATER QUALITY MODELLING OF LAGUNA LAKE, PHILIPPINES, accessed on March 27, 2026, https://pdfs.semanticscholar.org/16ee/8bacb432c6adce2e64b9824019ac1fab5735.pdf
45. Radiometric Correction of Landsat-8 and Sentinel-2A Scenes Using Drone Imagery in Synergy with Field Spectroradiometry - MDPI, accessed on March 27, 2026, https://www.mdpi.com/2072-4292/10/11/1687
46. Assessment of cross-sensor vegetation index compatibility between VIIRS and MODIS using near-coincident observations - SPIE Digital Library, accessed on March 27, 2026, https://www.spiedigitallibrary.org/journals/journal-of-applied-remote-sensing/volume-12/issue-4/045004/Assessment-of-cross-sensor-vegetation-index-compatibility-between-VIIRS-and/10.1117/1.JRS.12.045004.full
47. Spatial and Spectral Translation of Landsat 8 to Sentinel-2 Using Conditional Generative Adversarial Networks - MDPI, accessed on March 27, 2026, https://www.mdpi.com/2072-4292/15/23/5502
48. Regional Forest Wildfire Mapping Through Integration of Sentinel-2 and Landsat 8 Data in Google Earth Engine with Semi-Automatic Training Sample Generation - MDPI, accessed on March 27, 2026, https://www.mdpi.com/2072-4292/17/24/4038
49. Cross-Sensors Comparison of Popular Vegetation Indexes From Landsat TM, ETM+, OLI, and Sentinel MSI for Time-Series Analysis Over Europe - IEEE Xplore, accessed on March 27, 2026, https://ieeexplore.ieee.org/iel7/36/10354519/10360186.pdf
50. Multi-sensor anomalous change detection in remote sensing imagery - SPIE Digital Library, accessed on March 27, 2026, https://www.spiedigitallibrary.org/journals/journal-of-applied-remote-sensing/volume-15/issue-4/042411/Multi-sensor-anomalous-change-detection-in-remote-sensing-imagery/10.1117/1.JRS.15.042411.full
51. A Novel Spectral Vegetation Index for Improved Detection of Soybean Cyst Nematode (SCN) Infestation Using Hyperspectral Data - MDPI, accessed on March 27, 2026, https://www.mdpi.com/2673-7655/5/5/58
52. Removing an intersubject variance component in a general linear model improves multiway factoring of event-related spectral perturbations in group EEG studies - UT Dallas Research Labs, accessed on March 27, 2026, https://labs.utdallas.edu/memory/files/2020/06/spence-intersubject-variance.pdf
53. Amfiteatru Economic Journal, accessed on March 27, 2026,

https://www.amfiteatrueconomic.ro/temp/Journal_no_37.pdf
54. Cow Bayou Tidal (Segment 0511) Use Attainability Analysis - Texas Parks and Wildlife, accessed on March 27, 2026, https://tpwd.texas.gov/publications/pwdpubs/media/pwd_rp_v3400_1674.pdf
55. Applications Using Claims Dat MASSACHUSETTS INSTITUTE - DSpace@MIT, accessed on March 27, 2026, http://dspace.mit.edu/bitstream/handle/1721.1/45946/321051813-MIT.pdf?sequence=2
56. Sustainable Development and Informatics: Emerging Issues - Lincoln Research and Publishing Limited, accessed on March 27, 2026, https://www.lincolnrpl.org/pdf/ISBN%20Sustainable%20Development.pdf
57. Dedysh, SN - Royal Netherlands Academy of Arts and Sciences (KNAW), accessed on March 27, 2026, https://pure.knaw.nl/ws/files/459350/5476_Bodelier.pdf
58. Land-Use–Land Cover Changes in the Urban River's Buffer Zone and Variability of Discharge, Water, and Sediment Quality—A Case of Urban Catchment of the Ngerengere River in Tanzania - ResearchGate, accessed on March 27, 2026, https://www.researchgate.net/publication/381150070_Land-Use-Land_Cover_Changes_in_the_Urban_River's_Buffer_Zone_and_Variability_of_Discharge_Water_and_Sediment_Quality-A_Case_of_Urban_Catchment_of_the_Ngerengere_River_in_Tanzania
59. remote sensing and gis application in surface water quality research in the chepinska river basin (western - INTERNATIONAL JOURNAL CONSERVATION SCIENCE, accessed on March 27, 2026, https://ijcs.ro/public/IJCS-25-03_25_Tcherkezova.pdf